# Dynamic Market Exposure Using Adaptive Volatility and Drawdown Regimes

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## Introduction and Motivation

Financial markets exhibit persistent regime shifts characterized by alternating phases of stability, directional trends, and volatility-driven drawdowns. While buy-and-hold strategies may perform well in the long run, especially during a bull trend, they can suffer from significant drawdowns during volatile regimes which in turn leads to liquidity risks and risk-tolerance shifts. This motivated the development of a strategy that dynamically adjusts exposure based on market conditions.

The initial inspiration came from empirical observations when analyzing Meta Platforms (META), which experienced multiple sharp drawdowns and recoveries over the past decade, including the 2018 Cambridge Analytica decline, the 2020 COVID-19 crash, and the 2022 technology-sector selloff followed by the strong AI-driven recovery in 2023. From these incidents, it was possible to conclude that deep drawdowns can create recovery opportunities (as seen in the 2023 recovery), but exiting too early during sustained and steady trends can lead to missed upside potential.

The objective is therefore to combine mean-reversion (drawdown recovery) and trend-following (volatility regime) into a single adaptive strategy that adjusts how much capital is invested depending on market conditions rather than always remaining fully invested or fully out of the market. With the help of rolling historical volatility, exposure is reduced during high-volatility periods to mitigate losses, and increased during stable conditions to ride sustained trends. Instead of fixed thresholds, percentile-based adaptive thresholds normalized over a financial year are used to generalize across assets and market conditions and to avoid overfitting.

Historical daily data is obtained using the Python `yfinance` library. While META is used as the primary motivation, the strategy is tested across SPY, JPMorgan Chase, Exxon Mobil, and NVIDIA to evaluate robustness and adaptability across different sectors and volatility settings. The primary objective is not necessarily to maximize returns, but rather to improve risk-adjusted performance by reducing drawdowns and increasing the Sharpe ratio.

# Trading Strategy

The strategy is built around volatility, drawdown, and price position within the historical range.

Daily returns are defined as:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \qquad \begin{cases} r_t : \text{daily return} \\ P_t : \text{current price} \\ P_{t-1} : \text{previous price} \end{cases}$$

Rolling 30-day annualized volatility is computed as:

$$\sigma_t = \mathrm{Std}(r_{t-29:t}) \times \sqrt{252} \qquad \begin{cases} \sigma_t : \text{rolling volatility} \\ \mathrm{Std} : \text{standard deviation} \\ 252 : \text{trading days/year} \end{cases}$$

Volatility serves as the primary regime indicator. Normalization throughout the entire financial year (on average, 252 trading days per financial year) is ensured and thresholds are defined using rolling 252-day percentiles: the 85th percentile represents high volatility, while the 40th percentile represents low volatility.

Drawdown is measured relative to the rolling 252-day maximum. This captures downside deviation from recent highs and helps identify stressed market conditions:

$$DD_t = \frac{P_t - \mathrm{RollingMax}_t}{\mathrm{RollingMax}_t} \qquad \begin{cases} DD_t : \text{drawdown} \\ P_t : \text{current price} \\ \mathrm{RollingMax}_t : \text{252-day high} \end{cases}$$

Along with drawdown, the strategy uses a range position parameter which, with the help of the rolling 252-day peak and trough, gives the position of the current price in fractional terms, with values near 1 indicating strong upward positioning and values near 0 indicating proximity to yearly lows.

$$RP_t = \frac{P_t - \mathrm{RollingMin}_t}{\mathrm{RollingMax}_t - \mathrm{RollingMin}_t} \qquad \begin{cases} RP_t : \text{range position} \\ \mathrm{RollingMin}_t : \text{252-day low} \\ \mathrm{RollingMax}_t : \text{252-day high} \end{cases}$$

# Regime Classification and Exposure Logic

The strategy operates under three volatility regimes:

## 1. High Volatility Regime

When volatility is greater than the 85th percentile, market conditions are considered unstable. Exposure is reduced to 30% (meaning only 30% of one's capital is subject to market risks while the rest may be in cash or stored in other forms) to protect capital during unpredictable stress regimes.

## 2. Low Volatility Trend Regime

When volatility is less than the 40th percentile, markets are assumed to be stable. If the range position exceeds 0.60 and the slope is positive, indicating strong upward momentum, the strategy allocates 100% exposure. Otherwise, exposure is set to 60% to stay invested without taking excessive risks.

## 3. Moderate Volatility Mean-Reversion Regime

When volatility lies between the upper and lower thresholds, the strategy evaluates potential recovery setups. If drawdown falls below its adaptive historical threshold ($DD\_Q15$), indicating unusually deep stress, and the range position remains below 0.60, the strategy assumes the asset is in an oversold condition and increases exposure to 100%. Otherwise, exposure is maintained at 60% to ensure continued participation without overcommitting during uncertain conditions.

# Portfolio Construction

Final strategy returns are computed as:

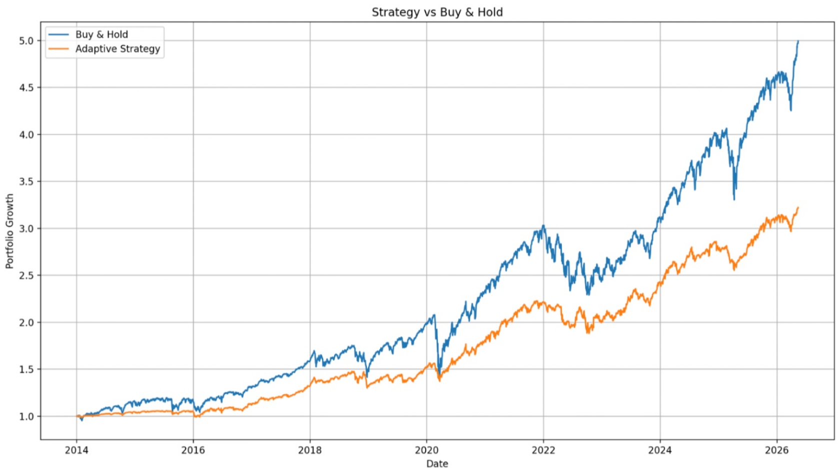
$$\text{StrategyReturn}_t = \text{Exposure}_{t-1} \times r_t \qquad \begin{cases} \text{StrategyReturn}_t : \text{strategy return} \\ \text{Exposure}_{t-1} : \text{capital invested} \\ r_t : \text{daily return} \end{cases}$$

This ensures capital allocation dynamically responds to prior-day market conditions, enabling gradual adjustment of risk exposure rather than abrupt shifts.

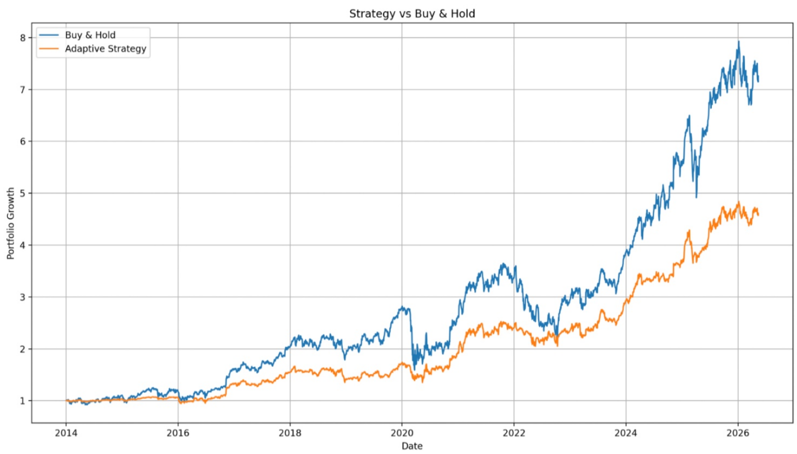
# Results and Backtesting

The strategy is tested on SPY, JPM, and META to evaluate performance across different market conditions. SPY shows the most stable results due to its smoother trend and lower volatility, which allows the regime-based exposure system to function effectively. JPM performs moderately well, capturing both trend-following and mean-reversion phases with some impact from higher volatility. META shows weaker performance due to high volatility and frequent regime shifts, which reduces the reliability of exposure decisions.

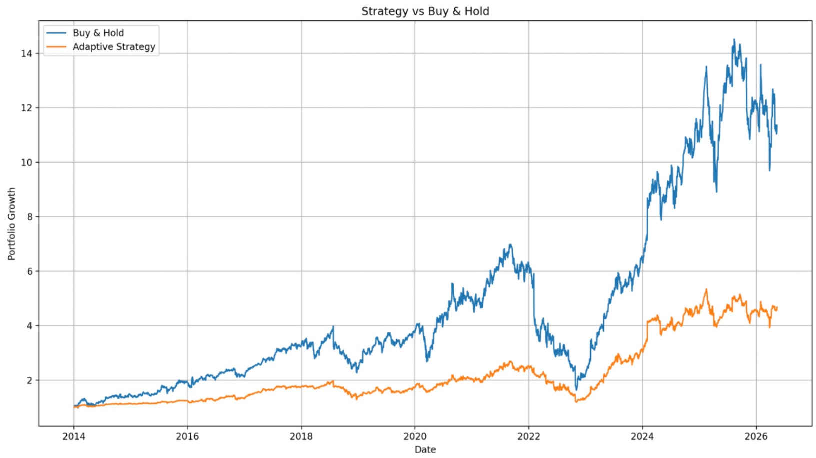
| Asset | Strategy Return | Strategy Max DD | Strategy Sharpe | Buy & Hold Return | Buy & Hold Max DD | Buy & Hold Sharpe |
|---|---|---|---|---|---|---|
| SPY | 222.25% | -15.59% | 1.00 | 399.60% | -33.72% | 0.84 |
| JPM | 357.54% | -22.41% | 0.85 | 615.17% | -43.63% | 0.74 |
| META | 367.53% | -56.16% | 0.64 | 1035.93% | -76.74% | 0.71 |




(a) Strategy vs Buy-and-Hold performance for SPY. The strategy exhibits smoother growth with significantly reduced drawdowns compared to the buy-and-hold benchmark.




(b) Strategy vs Buy-and-Hold performance for JPMorgan Chase. The adaptive exposure system captures both recovery and trend-following phases while reducing downside risk.




(a) Strategy vs Buy-and-Hold performance for Meta Platforms. High volatility and rapid regime shifts reduce the consistency of decisions made by the strategy when compared with SPY and JPM.

# Discussion, Limitations, and Robustness

The strategy demonstrated noticeably different behavior across asset classes, with the strongest performance observed on the S&P 500 ETF, SPY, and comparatively weaker performance on hyper-growth equities such as META. This divergence highlights an important characteristic of regime-based exposure systems: the effectiveness of the strategy heavily depends upon the type of asset and the market behavior.

SPY performed better because it is a broad market index with more stable long-term movement and volatility than individual equities. Since SPY represents a diversified basket of companies, risks concerned with the company specifically are naturally diluted. As a result, volatility is much more consistent and predictable which helps significantly with the strategy's adaptive exposure mechanism, which increases capital allocation during stable upward regimes and reduces exposure during periods of high volatility and drawdown stress. On the contrary, the strategy is not able to accurately read the market in cases of individual equities like META which constantly change between high and low volatility, often causing false signals. In addition, equity indices such as SPY have historically been known to show upward trends over long periods of time due to economic growth, allowing trend-following behavior to remain viable.

The strategy also appeared to work better on cyclical or macro-sensitive assets, since these assets move through clearer market phases such as expansion, correction, recovery, and consolidation. Volatility clustering, where periods of high volatility are followed by continued high volatility and periods of low volatility are followed by continued low volatility, also played a major role. Because the strategy relies on volatility, drawdowns, and range positioning, assets with more structured behavior produced more reliable signals while allowing the strategy to maintain higher exposure during stable conditions and reduce downside risk during stressed environments.

On the other hand, hyper-growth equities such as META presented greater challenges. These stocks can experience sudden large price movements due to earnings releases, market sentiment, or company-specific news. Because of this, past volatility is not always a reliable indicator of future price movement. In some situations, the strategy reduced exposure too early during temporary volatility spikes and missed strong recoveries afterward. In other cases, exposure remained high before sharp declines occurred due to unpredictable price jumps. This illustrates an important limitation of volatility-based systems which is that they generally respond well to structured market stress, but struggle against sudden information shocks and highly nonlinear price behavior.

The results also show an important tradeoff between risk and return. The strategy was not designed to maximize returns during aggressive bull markets. Instead, its main goal was to reduce drawdowns and create more stable performance over time, which would be particularly valuable during bear markets. Because of this, there were periods where the benchmark outperformed the strategy during bullish market episodes, since the strategy sometimes operated at lower exposure levels. However, this was balanced by improved downside protection during difficult market conditions and, from a risk management perspective, a smoother return profile may be preferable for long-term investing compared to maximizing returns alone. This can be clearly observed in the strategy's maximum drawdown and Sharpe ratio statistics.

Another important consideration is the possibility of overfitting. Since the strategy uses adaptive thresholds, rolling volatility estimates, and exposure scaling rules, there is always a risk that parameter choices unintentionally become tailored to historical data. To reduce this concern, the strategy was tested across multiple asset types and percentile-based thresholds were used instead of constant values. The fact that performance characteristics remained consistent across several datasets provides some evidence of robustness.

Future improvements to the strategy include making the exposure levels adaptive instead of using fixed allocations such as 30%, 60%, and 100%, which could allow smoother responses to changing market conditions. Another improvement could involve dynamically adjusting the volatility lookback period rather than using a fixed window, since shorter windows may react faster during volatile periods while longer windows may help reduce noise in stable markets.

Overall, the results suggest that the strategy is most effective in assets characterized by persistent trends, gradual volatility transitions, and relatively stable market structure. It performs less effectively in environments such as fast-moving hyper-growth equities. Even with these limitations, the results suggest that adaptive exposure management can improve downside control while still allowing participation in long-term market trends. The study also shows that understanding asset behavior and market structure is important when designing systematic trading strategies.

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